

claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_ee69079d-571\agent-a22fa4556d12824dc.jsonl`  
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```

Transcript

1. user (2026-06-12T14:55:39.436Z)

Repo: C:\Users\avidu\Projects\badminton-highlight-indexer (branch docs/pmf-field-signals ? uncommitted working-tree changes included; read the files as they are on disk).
Files under review (read all in full):
- docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md (NEW - the recruiting flyer the owner hands to CANDIDATES; the owner recruits personally)
- docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md (reframed - now the hired associate's visit PLAYBOOK: coach script Q1-Q7, demo-after-Q6 rule, student questions, ground rules, Phase-2 record-and-show-back)
- docs/FIELD_VISIT_ANSWER_SHEET.md (per-visit form incl. new demo-reaction + student blocks)
- docs/PMF_FIELD_SIGNALS.md (the pre-registered decision layer; demo/student data = product feedback, NOT verdict signals)
Grounding (read as needed): docs/PMF_MARKET_RESEARCH.md, docs/UI_PROPOSAL.md + docs/UI_ALPHA_EXECUTION_PLAN.md + docs/HOSTING_PLAN.md (what the UI/hosting can honestly claim today), CLAUDE.md guardrails (minors exclusion, consent).

Owner's spec (verbatim intent): hire a person who visits academies, talks to coaches AND students, to understand how the project helps them and gather improvement feedback - full stop. Stretch goal: that person helps record videos, we run the pipeline, show the academy a sample result ("how it will look for them"). UI will be ready by then; hosting plan exposes the service to a few allowlisted people. The flyer must let a reasonable (non-technical) person understand the gig; the owner recruits personally.

LENS: honesty + internal consistency across the 4 kit docs and the repo's actual state. Find REAL flaws:

- Claims about the product/UI/hosting that overstate today's reality (check UI_ALPHA_EXECUTION_PLAN status, HOSTING_PLAN) ? e.g. "a private web app is opening to a small set of invited academies": is that supportable as a near-term claim per the docs, or does it need softening? The owner says UI "will be ready by then" ? is the flyer's tense honest?
- Contradictions between the recruiting flyer and the playbook (duration, pay structure, who they talk to, demo rules, ground rules) ? a hired candidate reads BOTH.
- Guardrail compliance: minors (students at academies are often minors ? does the talking-to-students flow respect that? recording exclusion intact?), consent, contact privacy, the no-promises rule vs showing demos and inviting academies to a web app.
- Does the demo-after-Q6 rule stay consistent everywhere (playbook, sheet, signals doc)? Does anything still say the associate doesn't carry the demo?

2. user (2026-06-12T14:55:45.512Z)

```
1      # Field & Product Outreach Associate ? KhelSutra (Sports-Tech, Bengaluru)  
2  
3      **Help a Bengaluru sports-AI product meet its first real users.**  
4  
5      | | |  
6      |---|---|
```

7 | ****Duration**** | ~3?4 weeks, part-time (flexible; mostly afternoons/evenings when
academies are active) |
8 | ****Location**** | Bengaluru ? in-person visits to badminton academies |
9 | ****Pay**** | ? ___ per completed academy visit + travel reimbursement; recording sessions
(stretch goal below) paid extra ? ? ___ per session |
10 | ****Reports to**** | Avi (phone/WhatsApp: ___) |

11
12 ---

13
14 ## The product, in plain words
15

16 ****KhelSutra**** turns ordinary practice/match video from a badminton academy into
17 ready-to-use pieces: it automatically finds ****every rally**** in the recording, cuts
18 ****highlight reels****, and produces a simple ****match report****. It works on video from
19 a phone on a tripod ? no special cameras, no hardware to install. A private web
20 app is opening to a small set of invited academies.

21
22 ## The job, in one paragraph
23

24 Visit good badminton academies across Bengaluru and have honest conversations with
25 ****coaches AND students****: how do they work with video today, would something like
26 this actually help them, and ****what would make it genuinely useful for them**** ?
27 including what's missing or wrong with the idea. You work from a provided
28 conversation guide, write down what people ****actually say**** (the negative answers
29 matter most), and debrief with Avi after each visit. ****You are gathering feedback,**
30 **not selling.**** No tech background needed.

31
32 ## The stretch goal (extra pay)
33

34 For academies that are interested, you coordinate a simple recording session ?
35 a phone on a tripod, one-page checklist provided. We run that video through
36 KhelSutra, and you go back and show them ****their own court, their own players, as**
37 **highlight reels and a report****: ****"this is what it looks like for you."*** Then you
38 write down exactly what they say. There is no stronger feedback than that moment.

39
40 ## What you'll actually do
41

- 42 - Visit ****10?15 academies**** (intro + conversation, ~20?30 minutes each).
- 43 - Talk to the ****owner/head coach**** first, and ? with their permission ? a few
- 44 ****senior or adult students**** about whether they'd watch/share their own rally
- 45 clips and what they'd want from them.
- 46 - Fill ****one structured answer sheet per visit**** (we provide it) ? their words,
- 47 not yours, including the most negative thing said.
- 48 - Show a short sample reel on your phone ****only at the point the guide says to****
- 49 (after the open questions ? it changes people's answers if shown early).
- 50 - Debrief with Avi after each visit (WhatsApp/call). ****Your first two visits are**
- 51 **done together with him****, so you're never improvising.
- 52 - ****Stretch:**** coordinate recording sessions and present the sample results back.

53
54 ## You're a good fit if
55

- 56 - You're comfortable walking into an academy and starting a conversation with a
- 57 stranger ? and just as comfortable hearing "no, this is useless to us."
- 58 - You can capture what people said ****accurately**** ? quotes, not summaries.
- 59 - Kannada / Hindi / English, as the conversation needs (you choose the mix).
- 60 - You play or follow a racket sport (helps a lot; not required).
- 61 - No technical skills needed. Reliability matters more than anything else.

62
63 ## What you get
64

- 65 - Paid per completed honest visit ? ****not**** per positive answer. A well-documented
- 66 "they hated it, here's why" is a fully paid, fully successful visit.
- 67 - Ground-floor view of a real sports-AI product finding its market; your field
- 68 notes directly shape what gets built next.
- 69 - Flexible hours, short commitment, clear scope.

```

70
71     ## Ground rules (the short version ? full version provided on joining)
72
73     1. Honesty over positivity. Never oversell, never promise features, prices, or dates.
74     2. **No recording of any person without written consent** ? and sessions involving
75         **minors are never recorded** for our use, period. In regular visits you record
76         nothing at all: notes only.
77     3. People's contact details stay between you and Avi.
78
79     ## How to apply
80
81     Message Avi at ____ with ***"FIELD"**, one line about a sport you follow, and one
82     line about a time you had to get honest feedback out of someone. First reply gets
83     a 15-minute call.
84

```

3. user (2026-06-12T14:55:46.624Z)

```

1     # Field Visit Playbook ? academy conversations (KhelSutra)
2
3     > **Recruiting?** Hand candidates
4     [ `PRODUCT_FIELD_ASSOCIATE_FLYER.md` ] (PRODUCT_FIELD_ASSOCIATE_FLYER.md)
5     > ? that's the role description. **This document is the working playbook the hired
6     > associate gets on day one:** the conversation script, the ground rules, and the
7     > Phase-2 (recording / show-back) procedure.
8
9     **Duration:** ~3?4 weeks, part-time (flexible hours, mostly afternoons/evenings when
10    academies are active)
11    **Location:** Bengaluru ? in-person visits to badminton academies
12    **Compensation:** ? ____ per completed academy visit + travel reimbursement; Phase-2
13    sessions paid extra *(discussed individually)*
14    **Reports to:** Avi (phone/WhatsApp: ____ )
15
16    ---
17
18    ## The gig in one paragraph
19
20    KhelSutra takes ordinary practice/match video from a badminton academy and automatically
21    finds every rally, builds
22    highlight reels, and generates match reports; a private web app is opening to a few
23    invited academies. We need honest,
24    on-the-ground feedback from real academies ? **how they work with video, whether this
25    actually helps them, and what
26    would make it genuinely useful**. **Your job: visit 10?15 good academies, have a
27    friendly 20?30 minute conversation
28    with the owner/head coach (and, with permission, a few senior/adult students) using the
29    script below, and write down
30    what they actually say.** No selling. No tech knowledge needed. For interested academies
31    there's a Phase-2 follow-up
32    (a recording session ? then you return and show them THEIR OWN footage as highlights)
33    with additional compensation.
34
35    ---
36
37    ## Phase 1 ? The conversation (what you actually do)
38
39    **Who to talk to:** the academy owner or head coach ? the person who decides what the
40    academy spends money on.
41    If you get a junior coach, be friendly, gather what you can, and ask when the owner is
42    available.
43
44    **Opening (keep it this honest):**
45    > "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton
46    academies ? it takes normal practice
47    > video and automatically cuts it into rallies and highlights. He asked me to talk to a

```

few serious academies to

35 > understand how you work with video today. Could I ask you a few questions? Takes 15 minutes, and we're not selling

36 > anything today."

37

38 **The questions (ask in this order; the starred ones matter most):**

39

40 1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired person?

41 2. ? **Do you have recorded footage sitting around that nobody has time to watch?** (multi-court hall recordings,

42 tournament days, CCTV) ? roughly how much?

43 3. Who would watch rally clips or highlights if they existed ? coaches for training? players? parents?

44 4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory, Stupa, BadPro+, just

45 YouTube/phone.) What do you like / what's missing in it?

46 5. ? **If you could hand over a full session's video and get back every rally cut out + a highlight reel + a simple

47 report ? would you pay per match for that? What would that be worth to you, per match?** Then **stop talking and

48 wait** ? let THEM name the number. If they genuinely won't after a pause and one "roughly?", you may offer the

49 ladder: "?100, ?300, ?500?" **On the sheet, tick whether their number came BEFORE or AFTER you offered the

50 ladder** ? it changes what the answer means. (Note their face, not just the words.)

51 6. Your hall has multiple courts running at once, right? Does that make video less useful today?

52

53 **? NOW (and only now) show the sample reel** ? a ~60-second real highlight reel we put on your phone. Say: "this is

54 what it produces ? every rally found and cut automatically." Then shut up and watch their face. Write down their

55 FIRST reaction verbatim, and ask one follow-up: **"what would this need to do for it to be useful for YOUR academy?"**

56 (Improvement asks, in their words, go on the sheet. Showing it earlier anchors every answer above ? never lead with it.)

57

58 7. Would you let us record one session here (we pay a small fee for the time) and come back showing what the tool

59 produces on YOUR court, your players? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

60

61 **Students (with the coach's permission ? senior/adult students only, 5 minutes):** ask 2?3 of them:

62 (a) **"Would you watch clips of your own rallies if they existed after each session?"

63 (b) **"Would you share them ?

64 with whom?" (c) **"What would you want them to show?" Capture their answers in the sheet's student block. Don't

65 collect students' contact details; if everyone available is a minor, skip the student questions ? note that instead.

66

66 **After each visit, fill ONE answer sheet ? the template is

[`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md)

67 (print one per visit):** academy name, location, # courts, person spoken to + role + contact, answers 1?7 (their

68 words, not yours), the most negative thing they said (required), exact quotes, your gut read, and whether they're a

69 Phase-2 candidate. *(How the answers get scored and what totals trigger which decision lives in

70 [`PMF\_FIELD\_SIGNALS.md`](PMF_FIELD_SIGNALS.md) ? that part is ours, not yours; just capture honestly.)*

71

72 ---

73

74 ## Ground rules & gotchas (read twice ? this is the difference between useful and
useless)
75
76 1. **You are researching, not selling.** If they ask "how much will it cost", say
"that's exactly what he's trying to
77 figure out ? what would feel fair to you?" Never quote a price yourself.
78 2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did
you LAST record a session, and
79 what happened to that file?" gets the truth. Prefer the second kind of question.
80 3. **Don't promise anything.** No features, no dates, no free trials. You may SHOW the
standard sample reel (it's
81 real output ? at the scripted moment only, after Q6) and you may offer the Phase-2
idea (a paid recording session
82 + free sample output of their own court), and only as "would you be open to it".
Nothing else.
83 4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a
reason is worth MORE to us
84 than five vague yeses. You're paid per completed honest visit, not per positive
answer.
85 5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.**
Ask what it does well, what it
86 doesn't, and what they pay. Spend extra time here.
87 6. **Multi-court is our favorite topic.** If they complain that video is useless because
"all the courts are in frame
88 and it's chaos" ? capture that complaint in their exact words. That's the problem we
solve.
89 7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ?
notes only. Don't photograph
90 the academy or players unless the owner offers.
91 8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back
to record.
92 9. **Confidentiality:** don't describe the tech beyond the opening line, don't name
other academies' answers, and
93 don't share the answer sheets with anyone except Avi.
94
95 ---
96
97 ## Phase 2 (only for academies that said yes ? separate briefing & pay)
98
99 You coordinate a recording session: we pay the academy a small fee, set up a
phone/camera on a tripod (we provide a
100 one-page capture checklist: behind the court, elevated, whole court in frame, landscape,
1080p/60fps, don't move it),
101 record 122 sessions ? then we process the footage through KhelSutra and **you return to
present the academy its own
102 sample output** (their court, their players, as a highlight reel + report, free) and
fill a short reaction sheet:
103 what they said when they saw it, what they'd change, whether they asked to keep using
it. That show-back conversation
104 is the single most valuable artifact this whole engagement produces. **Hard rules:**
written consent from the academy
105 AND the adult players being recorded; **sessions with minors are not recorded for our
library** ? adult sessions only;
106 the academy gets told plainly what the footage is for (building and demonstrating the
tool).
107
108 ---
109
110 ## Day one
111
112 Your first 2 visits are paired (you + Avi, or a live phone walkthrough) so you're never
improvising. Before visit 1:
113 read this playbook twice, have the sample reel saved offline on your phone, and carry
printed answer sheets
114 ([`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) ? one per academy).

4. user (2026-06-12T14:55:53.124Z)

```

1      # Field Visit Answer Sheet ? one per academy (fill the same day)
2
3      > Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
4      > is a *good* sheet. Companion script: `FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
5      > Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` §2.
6
7      ## Visit basics
8
9      | | |
10     |---|---|
11     | Date / time | |
12     | Academy name | |
13     | Area (neighbourhood) | |
14     | # courts in the hall | |
15     | Spoke to (name + role) | |
16     | Contact (phone/WhatsApp) ? *only if offered* | |
17     | How long the conversation ran | |
18
19     ## The seven questions ? their answers (verbatim where possible)
20
21     **Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?**
22
23     >
24
25     **Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?**
26
27     >
28
29     **Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?**
30
31     >
32
33     **Q4 ? What do they use today (Machaxi / VISIST / Game Theory / Stupa / BadPro+ / none)?
34     What's good, what's missing, what does it cost them?**
35
36     >
37
38     **Q5 ? ? Would they pay per match for rally cuts + reel + report? What number did
39     THEY say (?)? What was their face doing?**
40
41     Their number came (tick one): ? BEFORE the ladder (they volunteered it) .
42     ? AFTER I offered ?100/300/500 . ? no number / refused .
43     ? they framed it differently (per student / per month ? write it below)
44
45     >
46
47     **Q6 ? Multiple courts in frame ? does that make video useless today? Their words:**
48
49     >
50
51     **Demo (shown AFTER Q6, never earlier) ? their FIRST reaction to the sample reel,
52     verbatim. Then: "what would this need to do to be useful for YOUR academy?" ?
53     their improvement asks, in their words:**
54
55     Demo shown? ? yes . ? no (why: _____ )
56
57     >
58
59     **Q7 ? Open to a paid recording session on their court (we return showing THEIR
60     footage as highlights)? (yes / maybe / no ? and who to call)**
61

```

```

62      >
63
64      **Students (coach-permitted, senior/adult only ? skip if only minors available):
65      would they watch clips of their own rallies? share with whom? what should clips
66      show? Their words:**
67
68      >
69
70      ## Required: the most negative thing they said (verbatim)
71
72      >
73
74      ## Best quotes (anything vivid, in their words)
75
76      >
77
78      ## Scores ? OFFICE USE, leave blank (scored by Avi at transcription per
79      PMF_FIELD_SIGNALS.md §2; when torn, score lower)
80      | Hypothesis | Score |
81      |---|---|
82      | H1 recording behavior | GREEN / AMBER / RED |
83      | H2 dead footage | GREEN / AMBER / RED |
84      | H3 audience | GREEN / AMBER / RED |
85      | H4 whitespace | GREEN / AMBER / RED |
86      | H5 willingness to pay ? their ? number: _____ | GREEN / AMBER / RED |
87      | H6 multi-court pain | GREEN / AMBER / RED |
88      | H7 Phase-2 door | GREEN / AMBER / RED |
89
90      ## Competitor block (only if they named one)
91
92      | | |
93      |---|---|
94      | Product | |
95      | What it does well (their words) | |
96      | What it misses (their words) | |
97      | What they pay | |
98
99      ## Your gut read (circle): COLD · POLITE · CURIOUS · EAGER
100
101      ## Phase-2 candidate? (circle): YES ? call: _____ · MAYBE · NO
102
103      ## Anything else worth knowing (layout, lighting, camera positions you noticed, who
104      really decides)
105
106      >

```

5. user (2026-06-12T14:55:53.126Z)

```

1      # C13 Field Signals ? what academy visits must tell us (and when to change course)
2
3      **Status:** ACTIVE ? the decision layer for the C13 beachhead validation
4      (owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
5      visiting academies and reading signals on the ground*).
6      **Companions:** [ `PRODUCT_FIELD_ASSOCIATE_FLYER.md` ](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
7      (the recruiting flyer candidates see) ·
8      [ `FIELD_RESEARCH_ASSOCIATE_FLYER.md` ](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
9      (the hired associate's visit playbook: script + ground rules + Phase-2) ·
10     [ `FIELD_VISIT_ANSWER_SHEET.md` ](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
11     form) ·
12     [ `PMF_MARKET_RESEARCH.md` ](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
13
14     **Why this doc exists:** the flyer collects answers; this doc fixes ? **before the
15     first visit** ? what those answers *mean* and what totals trigger which decision.

```

15 Pre-registering the thresholds is the honesty mechanism: it stops us from reading
16 15 ambiguous conversations as whatever we hoped to hear.
17
18 ---
19
20 ## 1. The hypotheses under test (each maps to script questions)
21
22 | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
23 |---|---|---|---|
24 | H1 | ****Recording behavior exists**** ? academies already produce fixed-angle / CCTV /
multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
25 | H2 | ****Dead-footage pain**** ? recorded footage piles up unwatched (the indexing wedge)
| footage is small, or already gets watched | Q2 ? |
26 | H3 | ****An audience exists**** ? coach, player, or parent would actually consume rally
clips/reels | shrug; "who has time to watch clips" | Q3 |
27 | H4 | ****Whitespace**** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+ ? the
India set is field-sourced from the 2026-06-10 competitive audit; PMF \$4's web table predates
it) leave a gap we fit | "we use X, it's fine, fully solved" | Q4 |
28 | H5 | ****Willingness to pay per match**** at a price that clears unit economics | "should
be free" / <?100 everywhere | Q5 ? |
29 | H6 | ****Multi-court chaos is a felt problem**** (our moat: indexing messy BYO multi-court
footage) | multi-court video already useful to them | Q6 |
30 | H7 | ****Behavioral commitment**** ? they'll let us record on their court (Phase-2 door) |
polite no across the board | Q7 |
31
32 Unit-economics floor behind H5: serving COGS on the Gemini path is ~\$0.30?0.60 per
33 hour of video (? 25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
34 roughly 75?150 to process, so ****?300/match has real margin, ?100/match is**
35 **break-even-ish at retail**** ? that's why the script's anchor ladder is ?100/300/500.
36
37 ## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
38
39 ****Scoring is done by the owner (or a Claude session) at transcription ? the**
40 **associate only captures answers**** (the sheet's score grid is marked OFFICE USE).
41 Score each hypothesis ****from what they said and showed, not what you inferred****.
42 When between two grades, take the lower one.
43
44 | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |
45 |---|---|---|---|
46 | ****H1 record**** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |
records occasionally (tournaments only) | doesn't record; no plans |
47 | ****H2 dead footage**** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",
tournament backlog) | some footage, partially watched | no backlog exists |
48 | ****H3 audience**** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I
review with students Mondays") | "could be useful, maybe" | no consumer identified |
49 | ****H4 whitespace**** | no tool today, or uses one and volunteers a gap we cover (multi-
court, BYO footage, price) | uses a competitor, mildly satisfied | competitor solves it,
satisfied, sticky |
50 | ****H5 WTP**** | names ??300/match ****BEFORE** the ladder was offered** (the sheet's tick-box
records this), or proposes a budget framing of their own whose per-match equivalent is ??300
(see counting rule below) | names a number only ****AFTER**** the ladder (anchored ? even
?300/?500), or volunteers ~?100 | refuses to name a number / "should be free" / volunteers <?100
|
51 | ****H6 multi-court**** | complains about it in their own words before/without prompting |
agrees when asked | multi-court footage already works for them |
52 | ****H7 Phase-2**** | yes + gives a name/number/slot | "maybe, come back" | no |
53
54 ****The H5 counting rule ("H5 at ?300+")****: a visit counts toward the "?300+" tally
55 only when the ? number (a) is ?300 ****and**** (b) came ***before*** the ladder was offered
56 ? an anchored ?300 is the modal polite answer to a read-aloud 100/300/500 ladder
57 and proves nothing, so it scores AMBER and does ****not**** count. Budget-framed
58 answers convert at transcription: their monthly ? ÷ the matches/month they cited
59 (if they cited none, ÷4); the computed equivalent counts if ??300 and the
60 conversion is written into the rollup row. Stated WTP ? even clean ? remains
61 ***talk***: the confirming step before any PROCEED-triggered build is the fake-door /

62 paid-pilot test already pre-agreed in PMF §6 risk 2 and the C13 row.

63

64 ****Weighting rule (behavior > words):**** H7-GREEN and H2-GREEN are ***behavioral***

65 signals and count double in your gut-read; H3/H5 answers are ***talk*** until a Phase-2

66 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a

67 polite conversation, not a customer.

68

69 ****Disconfirming-evidence quota:**** every visit sheet must record at least one

70 verbatim ***negative*** quote (the thing they liked least / wouldn't pay for). If a

71 sheet has only positives, the visit gets re-done or its evidence discounted in the

72 rollup (****never the associate's pay**** ? they're paid per completed honest visit);

73 per the flyer's ground rule 4, a reasoned "no" is worth more than five vague yeses.

74

75 ****Sampling note (read the verdict with this in mind):**** the flyer sends the

76 associate to 10?15 ***good*** academies ? the verdict therefore reads "does the

77 beachhead work where it should work best?", not "what fraction of the market is

78 addressable". A PROCEED here licenses the next validation step, not a TAM claim;

79 the rollup's area/courts columns record what was actually sampled.

80

81 **## 3. Pre-registered decision rules**

82

83 ****Stopping rule (fixes the optional-stopping loophole):**** the campaign targets

84 ****n = 15**** completed owner/head-coach sheets and may end earlier ****only for**

85 **logistics**** (associate availability, academies exhausted) ? ****never because of how**

86 **the tallies look****. The 5-sheet check-ins below are progress reports, not verdict

87 opportunities; ****the verdict is computed exactly once, at the final n****, and the

88 rollup records what that n was and why the campaign stopped. The count thresholds

89 below are written for n = 15 and apply unchanged at any final n in 10?14 ? which

90 deliberately makes the ?-bars ***relatively harder*** at smaller n (biases away from a

91 premature PROCEED) and the <-bars ***relatively harder*** to trip at n = 15 (biases away

92 from a premature FALSIFIED). Both biases err toward "keep investigating," the safe

93 failure mode. Counts, not percentages ? at this sample size we're reading

94 direction, not estimating proportions. "GREEN" below means GREEN on that

95 hypothesis row.

96

97 | Verdict | Trigger (pre-registered) | What we then do |

98 |---|---|---|

99 | ****PROCEED ? beachhead validated**** | ?6 H2-GREEN ****and**** ?4 H5 at ?300+ (per the §2

counting rule ? anchored numbers don't count) ****and**** ?3 H7-GREEN | Run the camera pilot at the

2?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list

(UI_ALPHA_EXECUTION_PLAN Phase R); the ****fake-door/paid-pilot WTP test**** (PMF §6 risk 2) runs

before money is spent on build |

100 | ****PIVOT A ? "index the backlog" ? "record + index" service**** | <3 H2-GREEN ****and**** ?6

visits with H3 GREEN-or-AMBER ****and**** ?3 H5 at ?300+ | The pain isn't the archive, it's that

nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes

the product wedge, not the demo |

101 | ****PIVOT B ? partner/API motion**** | H4 RED at ?5 academies (incumbents entrenched and

liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever

owns the academy relationship¹ |

102 | ****PIVOT C ? payer is the parent, not the academy**** | ?4 visits where H5 is RED ****and****

the Q3 answer names parents/players as the watcher | Re-test WTP as B2C per-student reels

(?/reel to parents) with the academy as channel, not customer |

103 | ****FALSIFIED ? beachhead assumption fails**** | <3 academies record at all (H1) ****and**** 0

H7-GREEN | The "academies already record fixed-angle" premise (PMF §5 beachhead #1; the academy

arm of §6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton

Asia/Denmark) before building further |

104

105 ¹ The template (surfaced by the 2026-06-10 competitive audit, not yet in PMF §4's

106 web-sourced table): ****PB Vision**** sells pickleball video analysis B2C at ~\$99/yr

107 ***and*** as an ~\$8/hr-of-video partner API through facility platforms like

108 ****PodPlay**** ? the engine is ours, the academy relationship belongs to whoever

109 already owns it.

110

111 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide

112 explicitly with the owner ? but ****the tallies above get computed first****, so the

discussion starts from the pre-registered read, not from anecdotes.

****Competitor-intel side-channel (always-on):**** any visit where a competitor is named ? fill the answer sheet's competitor block (what it does, what it misses, what they pay). ?3 consistent reports about one product = update
`PMF\_MARKET\_RESEARCH.md` §4 with field-verified rows (today's table is web-sourced).

4. Mechanics

- One [`FIELD\_VISIT\_ANSWER\_SHEET.md`] (FIELD_VISIT_ANSWER_SHEET.md) per academy, same day as the visit, in the associate's words + verbatim quotes.
- The owner (or Claude session) transcribes each sheet into one row of the rollup table below (keep it in this doc or a sheet next to it); tallies refresh per 5 visits as ****progress checks only**** ? the verdict is computed once, at the final n, per the §3 stopping rule.
- Demo asset: the associate carries the real compiled reel (`output/MahadevpuraSingles\_highlights.mp4` , trimmed to ~60s) saved offline on their phone, shown ****only** at the scripted moment (after Q6, before Q7) **** ? *show, never promise***. Demo reactions and improvement asks are ****product feedback****, not scored signals: they feed the build backlog verbatim, while the H1?H7 verdict machinery stays anchored to the pre-demo answers (that's why the demo comes last; the sheet records whether it was shown). Showing it right before Q7 also makes the Phase-2 ask a reaction to something seen, not imagined ? a stronger behavioral read.
- Student answers (sheet's student block) likewise feed product feedback and the H3 (audience) gut-read; they don't enter the verdict tallies, and no student contact details are ever collected.

Rollup table (one row per completed visit)

	#	Academy	Area	Courts	Role	spoken to	H1	H2	H3	H4	H5 (?)	H6	H7
Gut Phase-2?	?												
	1												

Tally ? decision read-out (progress check per 5 visits; verdict ONCE at final n)

Metric	Count	Threshold in play
H2 GREEN (dead footage)	?	6 ? PROCEED leg; <3 ? PIVOT A leg
H5 at ?300+ (pre-ladder / converted only ? §2 counting rule)	?	4 ? PROCEED leg; ?3 ? PIVOT A leg
H7 GREEN (Phase-2 yes + contact)	?	3 ? PROCEED leg; 0 ? FALSIFIED leg
H3 GREEN-or-AMBER (audience exists)	?	6 ? PIVOT A leg
H4 RED (incumbent satisfied)	?	5 ? PIVOT B
H1 any recording	?	<3 ? FALSIFIED leg
H5 RED **and** Q3 names parents/players	?	4 ? PIVOT C

5. Guardrails that bind the field work

- ****Phase 1 records nothing**** ? notes only (flyer rule 7).
- ****Phase 2 consent rules are hard:**** written consent from the academy AND adult players; ****sessions with minors are never recorded for our library**** (training-pool guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
- Answer sheets contain personal contact details ? they stay between the associate and the owner; transcribe to the rollup without phone numbers.

6. assistant (2026-06-12T14:55:57.298Z)

Now the grounding docs.

7. user (2026-06-12T14:55:58.242Z)

1 # UI Alpha Execution Plan ? KhelSutra Guru

```

2
3      **Status:** ACTIVE · updated 2026-06-10 · **Linked:** `UI_PROPOSAL.md` ·
`HOSTING_PLAN.md` ·
4      `mockups/ui_alpha_mockup.html` · ADR-009 (`archives/decisions/DECISIONS.md`).
5      **Sequencing principle (owner-requested):** build and test **everything locally first**;  

6      deployment (DNS/tunnel/Pages/Access) starts only after the local end-to-end gate passes.
7
8      ## Decisions locked
9
10     | Date | Decision |
11     |---|---|
12     | 2026-06-10 | Product name **KhelSutra Guru** · domain **khelsutra.guru** (GoDaddy,  

3-yr + protection) |
13     | 2026-06-10 | Separate **private repo `khelsutra`** ("heart and brain"): `web/` SPA  

now, control plane at beta; backend enablers stay in the indexer repo |
14     | 2026-06-10 | **Local-first delivery** (this plan's phase structure) |
15     | open | Final "go" on PR-1 coding start (= #16 greenlight) · training opt-in default  

OFF · invite list · stack (React+Vite+TS+Tailwind stands unless owner objects) |
16
17     ## Master checklist
18
19     ### G0 ? Setup (now)
20     - [x] Name/domain/repo-split ratified; spelling confirmed via GoDaddy receipt
21     - [x] `khelsutra` repo **scaffolded** at `C:\Users\avidu\Projects\khelsutra` (README,  

.gitignore, web/, docs/)
22     - [x] **Owner:** git-init + private GitHub push ? done 2026-06-10:  

`https://github.com/avidullu/khelsutra` (private; git runs Windows-side per ADR-009)
23     - [x] **Owner:** explicit "go" for PR-1 ? given 2026-06-10
24     - [x] **Owner:** Gemini API key rotated ? 2026-06-10, owner-verified (confirm old-key  

deletion in AI Studio)
25
26
27     ### Phase L ? build & test locally (no cloud; ?6?? sessions)
28
29     **L1 ? backend enablers** (indexer repo · branch off master · ONE PR each, no stacking ·  

offline tests in every PR · GPU/real-video checks = owner):
30     - [x] **PR-1 · B1 async job model (#16):** `jobs` table (user_version 1?), `POST  

/api/process` ? 202 + `job_id`, `GET /api/jobs/{job_id}`, ThreadPoolExecutor worker,  

restart/crash state handling ? **MERGED 2026-06-11, PR #87** (CI green: 333 passed; machine-side  

real-video checks fold into L3)
31     - [x] **PR-2 · B2+B3 upload & download:** chunked upload (init ? sequential ?95 MB parts  

? complete ? MD5; caps + mp4/mov allowlist; per-user dirs deferred to PR-3) + streaming `GET  

/api/highlights/{video_id}/{name}` ? **MERGED 2026-06-11, PR #99** (also fixes static-UI  

`segment_ids` compile bug + adds browser download link). **Live E2E verified at merge:** 86 MB  

real video uploaded in 3 sequential parts and reassembled to MD5 byte-identity; out-of-order 409  

/ traversal 400 (handler) + 404 (router) / unknown-id 404 confirmed; compile of a real confirmed  

rally ? streamed ? download (3.9 MB reel).
32     - [ ] **PR-3 · B4+B7 identity & exposure:** `owner_email` column + scoping on every  

query; Access-JWT/header read with `DEV_USER_EMAIL` env fallback for local dev; CORS narrowed;  

rate limit; per-user quota
33     - [ ] **PR-4 · B6 correction loop:** `PATCH /api/segments/{id}` accept/reject/boundary-  

nudge ? `candidate_segments.human_label`/`notes` + provenance (who/when); retire  

`backend/api/static` (B5)
34
35     **L2 ? SPA** (khelsutra repo, `web/`; dev server proxies to `http://localhost:8000`):
36     - [ ] **M1 scaffold:** Vite + React + TS + Tailwind, router, typed API client, branding  

(KhelSutra Guru)
37     - [ ] **M2:** A2 Upload (chunked, guidelines checklist) + A3 Job Progress (poll)
38     - [ ] **M3:** A1 Matches + A4 Rally Review (player seek, accept/reject/nudge wired to  

PR-4)
39     - [ ] **M4:** A5 Highlights (compile + download) + A6 Report + engine-offline banner
40     - [ ] Unit tests for API client + review-state logic; `npm run build` clean
41
42     **L3 ? local end-to-end (owner's box, real match videos):**
43     - [ ] Full UI flow on localhost: upload ? job ? review/correct ? compile ? download
44     - [ ] `python run_all_tests.py` green; UI polling vs SQLite WAL sanity; WASB/CLI paths

```

```

untouched
45     - [ ] **GATE GL:** owner signs off local E2E ? only then deployment work begins
46
47     ### Phase D ? deployment (?2 sessions + ~1 hr owner)
48     - [ ] **D1 (owner):** Cloudflare account ? add khelsutra.guru ? GoDaddy nameserver
switch (HOSTING_PLAN $5) ? scoped API token ($6) ? Pages project connected to GitHub `khelsutra`
(build root `web/`)
49     - [ ] **D2 (Claude):** DNS records; `cloudflared` Windows service on the box ?
`api.khelsutra.guru`; Access apps + email-allowlist policies for `app` & `api`; SPA prod env
50     - [ ] **D3:** deployed smoke E2E (owner-only allowlist); go-live security checklist
(HOSTING_PLAN $7) all green
51     - [ ] **GATE GD:** checklist signed off
52
53     ### Phase R ? rollout
54     - [ ] Wave 1: 5 testers (C13/camera-pilot contacts) + feedback channel (WhatsApp group
or Google Form)
55     - [ ] 2-week retro ? punch list ? widen toward 25 ? beta planning (UI_PROPOSAL $4)
56     - Metrics: matches/week · % rallies corrected · median turnaround · verdict mix · week-2
tester retention
57
58     ## Working agreement
59     Claude (Cowork) designs, codes, writes tests + docs, and leaves a
`scratch/PRn_HANDOFF.md` per
60     work item. **Claude Code executes all git/PR mechanics** (branch, commit, push, PR,
conflict
61     fixes, merges after owner review) by reading that handoff ? the owner never copy-pastes
62     commands (ratified 2026-06-11 after PR-1/#88 friction; supersedes the command-block
pattern).
63     Cowork's sandbox cannot run git on mounts (ADR-009) and CI runs the offline suite on
every PR.
64     One PR per item; no stacking; never push to master directly. PowerShell note: no `&&`.
65
66     ## Risks & mitigations
67     Box uptime (offline banner, alpha-acceptable) · Gemini 429 burst-throttle (serial
workers,
68     `ai_max_workers=1`, 6-attempt backoff ? learned 2026-06-10) · CF free-tier 100 MB body
cap
69     (chunked upload design) · single-GPU box (alpha default = Gemini path; WASB jobs queue-
depth 1) ·
70     .guru renewal cost ?18.8k/3-yr (optional: transfer to Cloudflare Registrar at-cost after
the
71     60-day registration lock; domain-protection may need toggling for transfer).
72
73     ## Session map (default order)
74     S1?S2 PR-1 · S3 PR-2 · S4 PR-3 · S5 PR-4 · S6 M1+M2 · S7 M3 · S8 M4 · S9 L3 E2E + fixes
.
75     S10 D1?D3 · S11 wave-1 launch
76

```

8. user (2026-06-12T14:55:58.659Z)

```

1     # Hosting Plan ? khelsutra.guru
2
3     **Status:** PROPOSAL ? awaiting owner ratification.
4     **Date:** 2026-06-10 · **Constraints:** budget $10?25/mo · GCP credits available ·
DigitalOcean
5     acceptable · alpha backend = owner's GPU box behind a tunnel · 10?25 alpha testers ·
registrar = GoDaddy.
6     **Companion:** `docs/UI_PROPOSAL.md`.
7
8     ## 1. Recommendation in one line
9
10    **Alpha: free edge + your box** (Cloudflare DNS/Pages/Access/Tunnel ? FastAPI on the GPU
box) at
11    **$0/mo**, saving GCP credits; **Beta: GCP control plane + your box as pull-worker**

```

```

(Cloud Run +
12     GCS + Firebase Auth, credits-covered), keeping the edge where it's free.
13
14     Rationale for the Cloudflare pieces despite the GCP preference: GCP has no free
equivalent of the
15     Tunnel + Access combo (IAP requires a load balancer, ~$18/mo standing cost), and the
alpha needs
16     exactly three things GCP charges for but Cloudflare gives free: a secure tunnel to a
home box with
17     a custom domain, email-allowlist auth (free ?50 users ? fits 10?25 testers), and static
hosting.
18     Credits are better spent at beta on compute/storage, which is where GCP is excellent.
19
20     ## 2. Alpha topology ($0/mo)
21
22     ```
23     tester browser
24         ? https
25         ?
26     Cloudflare edge (free): DNS + TLS + Access (email-OTP allowlist)
27         ??? app.khelsutra.guru ? Cloudflare Pages (React SPA; auto-deploy from GitHub)
28         ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only `cloudflared` service on
the box)
29
30         ?
31         ?
32         FastAPI :8000 on the GPU box (Windows)
33         ? Gemini segmenter path (no GPU in request path)
34         ? async jobs (#16), SQLite WAL, output/ on disk
35
36     - **No inbound ports opened** on the home network; `cloudflared` dials out. Home IP
never exposed.
37     - **Auth:** Cloudflare Access in front of BOTH hostnames; backend trusts the
38     `Cf-Access-Authenticated-User-Email` header (verify the Access JWT for rigor) for per-
user scoping.
39     - **Upload limit:** free-tier proxy caps request bodies at **100 MB** ? SPA uploads in
?90 MB
40     chunks (UI_PROPOSAL B2). Match videos (1?4 GB) flow fine as chunks.
41     - **Box offline = service offline.** Acceptable for alpha; SPA shows a friendly
"processing engine
42     offline" banner (a `/api/health` ping). Beta removes this with the queue.
43     - **Costs:** Pages, Tunnel, Access, DNS = $0. Only the Gemini per-run cost
(~$0.05/match) and
44     domain renewal.
45
46     ## 3. Beta topology (GCP credits; ~$15?20/mo post-credits)
47
48     ```
49     browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru (Pages)
50                                     ??? api.khelsutra.guru ??? Cloud Run
(FastAPI control plane, scale-to-zero)
51                                     ? jobs
table (Cloud SQL Postgres)
52                                     ? signed
resumable URLs
53                                     ?
54                                     GCS buckets
(uploads / reels, lifecycle ? cold)
55                                     ? pull
jobs, outbound-only
56                                     GPU box = first
`byo_worker` (PLATFORM P3)
57     ```
58
59     - Uploads go **browser ? GCS directly** (resumable signed URLs) ? no proxy body limit,

```

```

no home
60     bandwidth for ingest; the box pulls only what it processes (WASB/owned-model jobs),
Cloud Run
61     handles Gemini-path jobs entirely in-cloud.
62     - Auth graduates to **Firebase Auth** (free; Google + email sign-in; India-friendly
phone OTP
63     later); Access allowlist retires.
64     - Post-credits estimate: Cloud Run ~$0?5 (free tier: 2 M req + 180 K vCPU-s/mo, scale-
to-zero) .
65     GCS 100 GB ? $2.60 + egress $0.12/GB (reel downloads ? watch this line) . Cloud SQL
smallest
66     ? $10 ? **~$15?20/mo**, inside budget. Egress lever if it grows: serve reels from
Cloudflare R2
67     (zero egress) while keeping everything else GCP.
68
69     ## 4. Alternatives considered
70
71     | Option | Monthly (alpha ? beta) | Pros | Cons |
72     |---|---|---|---|
73     | **Recommended hybrid** (CF edge + box ? +GCP) | **$0 ? ~$0 w/ credits** (~$15?20
after) | Cheapest at every phase; credits used where they matter; maps to PLATFORM P2/P3 | Two
vendors (CF + GCP) |
74     | **All-DigitalOcean** | ~$11 ? ~$17 ($6 droplet + Spaces $5: 250 GiB + **1 TiB egress
included**) | One simple vendor; Spaces' included egress is the best video-download deal; per-
second billing | No credits; no scale-to-zero; still needs a tunnel-equivalent to reach the box
(run `frp`/WireGuard on the droplet) |
75     | **All-GCP from day 1** | ~$18+ ? credits | Single vendor, credits immediately | LB+IAP
standing cost just to gate an alpha; no free tunnel to the home box; credits burn on plumbing |
76     | **All-Cloudflare** (Pages+Workers+R2) | ~$0 ? ~$5 | Zero egress on R2; cheapest
forever | FastAPI doesn't run on Workers (API stays on box/Cloud Run anyway); ignores GCP
credits |
77
78     ## 5. Domain setup ? GoDaddy ? Cloudflare DNS (one-time, ~15 min, you)
79
80     1. Create a free Cloudflare account ? **Add a domain** ? `khelsutra.guru` ? Free plan.
Cloudflare
81     imports existing records and shows **two nameservers** (e.g.
`xxx.ns.cloudflare.com`).
82     2. GoDaddy ? **My Products ? khelsutra.guru ? Manage DNS ? Nameservers ? Change ?
83     "I'll use my own nameservers"** ? paste the two Cloudflare nameservers ? Save.
Propagation:
84     minutes to 24 h. (Registration/renewal stays at GoDaddy; only DNS moves. Optional
later:
85     transfer the registration to Cloudflare Registrar for at-cost .guru renewals.)
86     3. Records (I manage after takeover): `app` ? Pages project; `api` ? `**Cloudflare API token** (you create: dash ? My Profile ? API Tokens) | Zone:DNS:Edit,
Account:Pages:Edit, Account:Access:Apps:Edit, Account:Tunnel:Edit ? scoped to this zone | DNS
records, Pages deploys (wrangler), Access policies, tunnel config |
94     | **GitHub repo** `khelsutra` (private) | Pages connects once (dashboard, you; build
root = `web/`) ? every push auto-deploys | I write the SPA in your Projects folder; commits
pushed = deployed |
95     | **On the box** (one-time, I hand you exact commands) | `cloudflared` installed as a
Windows service + `config.yml` | The tunnel itself |
96     | **GCP (beta only)** | A `khelsutra` project + service account (Cloud Run admin,
Storage admin, Cloud SQL admin) | Control-plane deploys |
97
98     Day-to-day flow after setup: I build ? commit ? push ? Pages deploys the SPA
automatically;

```

```
99     API/tunnel/DNS changes via the scoped token. Secrets (Gemini key, Access policy) never
live in the
100     repo; `.env` on the box + Cloudflare/GCP dashboards.
101
102     ## 7. Go-live security checklist (alpha)
103
104     Access allowlist ON for both hostnames ? CORS narrowed to `https://app.khelsutra.guru` ?
rate
105     limit + per-user upload quota ? `security.allowed_video_root` set ? upload size cap +
extension
106     allowlist ? **rotate the Gemini API key** (NEXT_STEPS housekeeping ? it was shared in
chat) ?
107     confirm no router port-forwards ? weekly backup: SQLite DB + `output/` ? GCS bucket
(rclone
108     scheduled task; credits).
109
110     ## 8. Decisions needed from owner
111
112     1. Ratify the **hybrid recommendation** (CF free edge now, GCP credits at beta)? Or
force
113         single-vendor (DO / GCP) despite the cost notes above?
114     2. OK to move **DNS to Cloudflare** (registration stays at GoDaddy)?
115     3. Beta reel storage: **GCS (credits)** vs **R2 (zero egress)** vs **DO Spaces (1 TiB
included)** ?
116         can defer until beta Drop 2.
117     4. Approve the access grants in S6 so I can run the setup end-to-end.
118
119     *Ratified choices get recorded in `docs/archives/decisions/DECISIONS.md`.*
120
```

9. user (2026-06-12T14:56:05.107Z)

```
1     # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3     > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4     > validate live before relying on them for investment/pricing decisions. This is the
5     > market-research-of-record behind the PMF section of the living
6     > [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker (S6). Product framing:
7     > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8     > self-recorded, **static-camera, full-court amateur** racket-sports video and
9     > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10    > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12    > **Confidence convention.** Figures from named industry reports, federations, or
13    > company pricing pages are tagged with source + year. Inferred items are tagged
14    > ***(estimate)** / ***(low confidence)***. Emerging-category market sizes vary widely
15    > between vendors; ranges are shown rather than false precision.
16
17    ---
18
19    ## 1. Executive summary
20
21    The product sits at the intersection of three favorable trends: (a) a fast-growing
22    **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
23    depending on vendor); (b) a structural shift to **amateur self-recording** on action
24    cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
25    segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
26    106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
27    2024, +45.8% YoY).
28
29    The central tension is the **static-camera constraint**: the tool works best on fixed,
30    full-court amateur footage and *fails* on broadcast/panning footage. This is both a
31    moat (most competitors target team sports / broadcast) and a filter on the addressable
32    user ? the customer must already set a camera on a tripod behind the court. That
33    behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not
```

the median casual player.

****Recommended beachhead (strongest single):**** ****badminton coaching academies and serious club/competitive players in India**** ? a large, rapidly formalizing academy ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing willingness to pay for coaching, an underserved tooling gap (incumbents are team-sports/broadcast), and a direct-but-immature competitor field. The product is ***native badminton-first*** ? a genuine edge here. Secondary: ****(2) US/global pickleball players & clubs**** (capital-rich, fastest-growing, but SwingVision already strong); ****(3) badminton academies/clubs in East/SE Asia + Denmark**** (highest density).

****Pricing**** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo) and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A software-only BYO-camera model undercuts the hardware players; realistic anchors ****(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.****

****Top risks:**** (1) amateurs won't reliably set up a tripod ? TAM collapses to coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia, China) are the most price-sensitive and hardest to bill at Western SaaS prices.

2. Market sizing & trends

2.1 Sports analytics / sports-video market

- **2025 size:**** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business Insights); other vendors USD 5.6?6.0B. ***(2025)***
- **Growth:**** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%); Future Market Insights ? ~USD 36.2B by 2035 (22.1%). ***(reports 2024?2025)***
- **Video analytics is the leading segment**** within sports analytics (Grand View / Fortune BI) ? the directly relevant sub-market.
- **Interpretation (estimate):**** the ***amateur/club auto-highlight*** slice is a small but fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption

- **2024 size:**** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). ***(2024?2025)***
- **Sports is the largest end-use segment**** of action cameras (GMIInsights, 2024).
- **GoPro ~47% share, DJI ~20%**** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- **Implication:**** the capture-side behavior the product depends on is already widespread and growing ? ***for the segments that record*** ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)
Badminton	220M+ play regularly; 160+ countries BWF est. via BadmintonBites (2023)	
Badminton ? China	~80M registered, up to ~100M (largest base) industry compilations (2023)	
Badminton ? India	20M+ recreational by 2023; academy boom + Khelo India industry compilations (2023)	
Badminton ? Indonesia	National sport; ~1M club players industry (est.)	
Badminton ? Denmark	100,000+ registered club members (Europe #1) industry (2023)	
Badminton ? Japan/Korea	Major school-club + workplace sport industry (2023)	
Tennis	106M players (2024)**; +25.6% over 5 yrs (87M in 2021) ITF Global Tennis Report (2021, 2024)	
Tennis ? Asia	~1/3 of players (33M of 87M) ITF (2021)	
Pickleball ? US	19.8M players (2024)**; +45.8% YoY, +311% over 3 yrs; ~6.2M core SFIA / Pickleheads / The Dink (2024)	
Pickleball ? US courts	~70,000 courts, +55% places to play YoY SFIA / Pickleheads	

(2024) |

94 | Pickleball ? global | IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim ****(low confidence)**** | various (2024?2025) |

95 | ****Table tennis**** | 30M+ players; 227 member associations | ITTF (current) |

96 | ****Cricket**** | 2.5B+ fans; 500M+ in India (mostly ***fans***, not self-recorders) | Cricket Rise / Nielsen (2024?2025) |

97

98 - ****Pickleball market value:**** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR

99 (Market.us/Custom Market Insights, 2024?2025).

100 - ****Badminton equipment market:**** ~USD 7.23B global (2023), Asia-Pacific dominant.

101

102 ****Geographic read:**** badminton density is overwhelmingly Asian + Denmark; tennis is broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket is huge as ***fandom*** but its amateur self-recording + static-full-court fit is weak

103 (large field, not a fixed-frame court). For ***this product's static-camera fit***,

104 badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap item, not a beachhead.

105

106

107

108

109 ---

110

111 **## 3. Customer segment evaluation**

112

113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit | B2C/B2B | Verdict |

114 |---|---|---|---|---|---|---|

115 | ****Individual amateur/enthusiast**** | Largest raw TAM (100s of millions) | ****Low**** ? competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | ****Mixed**** ? most won't set a tripod | B2C | Volume play, weak economics; use as funnel |

116 | ****Coaches & coaching academies**** | India academy boom; thousands across Asia | ****High**** ? already sell coaching; USD 300?1,500/yr plausible | Reachable via federations/distributors/social | ****Strong**** ? already film fixed-angle | B2B/B2B2C | ****Best fit**** ? buyer, recorder, payer aligned |

117 | ****Clubs & leagues**** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD 500?3,000/yr | Medium ? committee sales cycle | ****Strong**** | B2B/B2B2C | Strong secondary; longer cycle |

118 | ****Tournament/event organizers**** | Many small/regional events | Medium ? per-event/seasonal | Medium ? episodic | ****Strong**** | B2B | Good per-event upsell; lumpy |

119 | ****Schools & universities**** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | ****National/regional federations**** | Few, high-value | Medium-high; procurement-heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | ****Broadcasters / streamers**** | High value per deal | High | Low ? incumbents entrenched | ****Poor** ? panning = failure mode** | B2B | ****Avoid**** ? violates core constraint |

122

123 ****Key dynamics:**** the static-camera constraint is a ****B2B filter**** ? coaches/academies/

124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest

125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters

126 explicitly violate the constraint ? exclude.

127

128 ---

129

130 **## 4. Competitive & pricing benchmarks**

131

132 | Product | Category | Model | Price | Notes (year) |

133 |---|---|---|---|---|

134 | ****Veo (Cam 3)**** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; \$115M raised (2025) |

135 | ****Trace**** | Soccer auto-filming | ****Free camera + sub**** | USD 180/yr (Basic) ? 300 (Pro) | Camera leased free with sub (2025) |

136 | ****Pixellot (Air)**** | Auto camera/streaming | Hardware + sub | Camera ~USD 949 + USD 69?167/mo | Institutional/league; 10,000+ clubs (2025) |

137 | ****Spiideo**** | Auto-filming/analytics | Sub incl. camera | ~USD 3,800/yr (1 team) ? 6,100/yr (club) | Institutional budgets (2025) |

138 | ****Hudl**** | Team video analysis | SaaS tiers | From ~USD 400/yr | Broad team-sports incumbent (2026) |

139 | ****SwingVision**** | ****Tennis/pickleball AI, single iPhone**** | Software freemium | ****USD**
179.99/yr******; free 2 hrs/mo | Closest analog; 200k+ monthly users; \$10M+ raised (2025) |
140 | ****BadPro+**** | ****Badminton AI on phone**** | ****Pay-per-use, no sub**** | Per-analysis (? "<
one coaching session") | Direct badminton competitor; shot detection, heatmaps, auto-highlights
(2025) |
141 | Niche pickleball/badminton apps | Various | Mixed | Varies | Fragmented; few true
auto-highlight tools (2025) |
142
143 ****What buyers actually pay (synthesis):**** consumer software-only anchors at ~USD
144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,
145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses **pay-per-use** ?
146 signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity +
147 warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and
148 sells worldwide via SaaS. Realistic anchors ****(estimate):** B2C USD 60?180/yr; B2B
149 academy/club USD 300?1,500/yr****** per location. The differentiators (cheap offline
150 self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B****
151 value story more than a B2C one.
152
153 ---
154
155 **## 5. Recommended beachhead**
156
157 Ranked by ****WTP × participation density × existing recording behavior × static-camera**
158 **fit × competitive whitespace.****
159
160 **### #1 (strongest) ? Badminton coaching academies & serious players, ****India******
161 20M+ recreational players (2023); rapidly ****formalizing academy ecosystem**** + Khelo
162 India tailwind; academies already record fixed-angle; coaches are buyers who monetize
163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton
164 AI (BadPro+) is early with unproven pay-per-use. Product is **native badminton-first**.
165 ****Watch-outs:**** price sensitivity (need INR-tuned tiers); sell to the ****academy****, not
166 the individual.
167
168 **### #2 ? Serious pickleball players & clubs, ****US (then UK/Australia/Asia)******
169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;
170 strong recording/sharing culture; pickleball is on the roadmap. ****Watch-out:****
171 ****SwingVision already owns single-camera tennis+pickleball AI**** at USD 179.99/yr (200k+
172 monthly users) ? head-to-head differentiation, not greenfield.
173
174 **### #3 ? Badminton academies/clubs, ****East/SE Asia (China, Indonesia, Malaysia, Japan,****
Korea) + Denmark******
175 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea
176 school-club & workplace systems), mature club infra, fixed-court recording.
177 ****Watch-outs:**** China needs local hosting/partners + is regulatory-complex; SE Asia
178 price-sensitive; Denmark/EU small but high-WTP and easy to pilot.
179
180 ****Bottom line:**** lead with ****India badminton academies**** (whitespace + behavior + buyer
181 alignment), use ****enthusiast B2C as the funnel****, validate a ****club/academy B2B2C**
182 **motion****, then expand to ****pickleball-US**** (out-build SwingVision) and
183 ****badminton-Asia/Denmark**** (density).
184
185 ---
186
187 **## 6. Risks & open questions (what would invalidate the thesis)**
188
189 1. ****Tripod/static-camera adoption among amateurs.**** If serious amateurs won't set up a
190 fixed camera, B2C TAM collapses to coaches/clubs. **Validate:** observe 3?5 academies +
191 50?100 enthusiasts on whether they already record fixed-angle.
192 2. ****Free / near-free alternatives**** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193 B2C WTP. **Validate:** fake-door pricing at USD 60?180/yr.
194 3. ****Capture-quality variance = the failure mode**** (gym lighting, off-center angle,
195 occlusion, mixed resolutions). **Validate:** benchmark rally-detection P/R on 100+ real
196 amateur clips; define a "good-capture" spec + auto-reject (ties to
197 `VIDEO\_RECORDING\_GUIDELINES.md` + the observability reports).
198 4. ****Monetization mismatch in best-participation markets**** (India/Indonesia/China are

price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy INR licensing.

5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports). *Validate:* ship a badminton-quality advantage incumbents can't trivially match (offline self-improving detection, per-rally tagging).
6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/academy self-serve before institutional deals.
7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real purchasing criterion via 5?10 buyer interviews.

Cheapest next steps: (a) 10?15 discovery interviews with Indian badminton academy owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).

7. Sources

- Sports analytics market: Precedence Research; Fortune Business Insights; Grand View Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
 - <https://www.precedenceresearch.com/sports-analytics-market>
 - <https://www.fortunebusinessinsights.com/sports-analytics-market-102217>
 - <https://www.grandviewresearch.com/industry-analysis/sports-analytics-market>
 - <https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp>
 - <https://www.futuremarketinsights.com/reports/sports-analytics-market>
- Action camera market: <https://www.gminsights.com/industry-analysis/action-camera-market> (2024);
 - <https://www.precedenceresearch.com/action-camera-market> (2024);
 - <https://electroi.com/stats/action-camera-statistics/> (2024)
- Badminton participation/market: <https://badmintonbites.com/how-many-badminton-players-are-there-in-the-world/> ;
 - <https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-popular/> ;
 - <https://en.wikipedia.org/wiki/Badminton> ;
 - <https://www.marketgrowthreports.com/market-reports/badminton-market-110049> ;
 - <https://www.thesportsschool.com/top-10-badminton-academies-in-india/>
- Tennis: <https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-participation-hits-106-million-in-five-years/> ;
 - <https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/>
- Pickleball: <https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/> ;
 - <https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/> ;
 - <https://www.pickleheads.com/guides/pickleball-statistics> ;
 - <https://www.custommarketinsights.com/report/pickleball-market/> ;
 - <https://market.us/report/pickleball-market/>
- Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
- Cricket: <https://cricketrise.com/cricket-fans-in-india/> ;
 - <https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/>
- Competitors & pricing: <https://www.veo.com/en-us/pricing> ; <https://traceup.com/pricing> ;
 - <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
 - <https://swing.vision/subscribe> ;
 - <https://www.badproplusapp.com/>

10. user (2026-06-12T14:56:05.886Z)

M docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md
 M docs/FIELD_VISIT_ANSWER_SHEET.md
 M docs/NEXT_STEPS.md
 M docs/PMF_FIELD_SIGNALS.md
 ?? arial.ttf

```
?? artifacts/  
?? docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md  
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md  
?? scratch/PR1_BODY.md  
?? scratch/PR1_HANDOFF.md  
?? scratch/PR2_HANDOFF.md  
?? scratch/PR2_deferred_hardening_status.patch  
?? scratch/PR88_FIX.md  
docs/pmf-field-signals
```

11. assistant (2026-06-12T14:57:42.019Z)

Now let me verify a few factual claims ? the demo asset, the "match report" feature, and any stale references elsewhere (NEXT_STEPS.md is also modified on this branch).

12. user (2026-06-12T14:57:43.623Z)

```
Compressed Videos  
MahadevpuraSingles_highlights.mp4  
adarsh_avi_full_vs_human.csv  
adarsh_avi_vs_human.csv  
adarsh_baseline_gated.csv  
adarsh_baseline_segments.csv  
adarsh_gemini_fullvideo.csv  
adarsh_gemini_refined.csv  
adarsh_gemini_smoke.csv  
adarsh_min5s_segments.csv  
adarsh_tuned_segments.csv  
badminton_indexer.db  
e2e_reel.mp4  
e2e_server_log.txt  
gbaaddy_vs_human.csv  
human_lovo_manifest.json  
kushagra_1280.mp4  
kushagra_1280.report.json  
kushagra_1280.report.md  
kushagra_1280.summary.md  
kushagra_vs_human.csv  
largetestvideo_vs_human.csv  
local_calib_manifest.json  
local_rally_model.json  
mahadevpura_gemini_refined.csv  
mahadevpura_smoke_180s.mp4  
mahadevpura_smoke_180s.report.json  
mahadevpura_smoke_180s.report.md  
mahadevpura_smoke_180s.summary.md  
mahadevpura_vs_human.csv
```

13. user (2026-06-12T14:57:44.951Z)

```
3:**Purpose:** cold-start navigation map for the badminton-highlight-indexer ? a self-hosted  
video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a  
segmenter-agnostic eval/calibration stack, typed config, and per-video observability reporting.  
29:         [validation FAIL -> persist status="failed", return early; report STILL emitted in  
finally]  
35: -> finally: emit_indexer_report(...) @backend/reporting/__init__.py (? ALWAYS runs ?  
even on validation failure / exception; NEVER raises; grafts response["reports"])  
67: -> harness.evaluate -> EvalReport -> report_to_dict  
70:Calibration: calibrate_wasb.run -> calibration.{select_best, improvement_report,  
cross_validate} @backend/eval/calibration.py  
85:[Omitted long matching line]  
117:[Omitted long matching line]  
121:[Omitted long matching line]  
133:- `@backend/eval/calibration.py` ? overfit guard. ``:cross_validate` (within-video k-fold; ?
```

```

defaults `metric='recall'` deliberately), `::leave_one_video_out` (honest cross-video; needs ?2
labelled videos), `::improvement_report` (OPTIMISTIC ? selected on full GT), `::select_best`,
`::kfold_indices`, `::pct_improvement` (? from-zero -> `inf`), `::evaluate`. $ `PredictFn =
Callable[[Config], List[Interval]]` = the plug-in seam. ? `generalization_gap >~0.1` = overfit
alarm.
138:- `@backend/eval/harness.py` ? DB-pred scorer (no re-run).
`::STAGES=('candidates','final')`, `::get_predictions`
(candidates->`query_candidate_segments(detector=)`; final->`query_play_segments({})`; ? unknown
stage raises ValueError; reused by regression.py),
`::evaluate`->`::EvalReport`(`::StageReport`)->`::report_to_dict` (input contract for
batch.aggregate), `::format_report_text`.
146:[Omitted long matching line]
170:[Omitted long matching line]
179:### 2.8 Reporting / observability ? (per-video, soft-dep `obsreport`)
180:[Omitted long matching line]
181:[Omitted long matching line]
182:[Omitted long matching line]
183:[Omitted long matching line]
203:$ **jobs row** (`db.create_job` / `get_job` ? #16, PR-1): `{id:TEXT(uuid4.hex), video_path,
video_id?:TEXT(set once hashed), segmenter?, player_pool?:JSON list, max_frames?,
status:'queued'|'running'|'done'|'failed' (EXECUTION state ? pipeline verdict is
result["status"]), progress?:TEXT(stage label), error?:TEXT, result?:JSON(the former sync
/api/process response incl. reports), created_at, started_at?, finished_at?}`.
221:[Omitted long matching line]
248:- **Gemini free-tier / API-key dead path:** hybrids + gemini return `[]` (not error) if
`{PROVIDER}_API_KEY` missing OR WSL healthcheck fails ? a silent no-op. `analyze_video` also
returns `[]` on real failure -> empty != "no rallies". The reporting
`silent_provider_noop`/`ai_empty_vs_no_rallies` rules exist precisely to disambiguate this in
the report.
249:- **Reporting never breaks a run:** `emit_indexer_report` swallows all exceptions (designed
for `main.py` `finally:`) and is emitted even on validation failure. `obsreport` is a SOFT dep ?
absent -> no-op.
251:- **fps / frame_width must be PROBED not assumed:** `_probe_fps_width` / runner fallbacks
(30.0, 1280.0) silently mis-scale velocity thresholds; calibrate_wasb CLI bakes `59.94/1920`.
The report's `fps_fallback_used` flag surfaces this.
274:| Add a segmenter | subclass `@backend/pipeline/segmenters/base.py::BasePlaySegmenter`;
`segmenter_registry.register(...)` at module bottom; import in
`@backend/pipeline/segmenters/__init__.py`; if AI-based add to
`@backend/reporting/harvest.py::_AI_SEGMENTERS` |
279:| Add / change a report footgun rule | `@backend/reporting/spec.yaml::rules` (declarative; ?
`when.path` must match harvest's stage/metric/detail names); facts in
`@backend/reporting/harvest.py` |
280:| Tweak what the report records | `@backend/reporting/harvest.py::record_run` + stage
builders; emit point is `@backend/main.py` `finally:` ->
`@backend/reporting/__init__.py::emit_indexer_report` |
312:| `test_eval_harness.py` | `eval.harness` DB-pred fetch + evaluate + report_to_dict; also
`@run_eval.py::main`/`get_file_md5` (exit codes, `--scaffold`/`--json`) |
314:| `test_calibration.py` | `eval.calibration`
pct_improvement/kfold/select_best/cross_validate/LOVO/improvement_report |
329:| `test_reporting_harvest.py` | `reporting` (`emit_indexer_report`, `harvest.record_run`); ?
`importorskip("obsreport")`; all 8 footgun rules + plain-dict-OR-AppConfig +
`report.enabled=False` |
330:| `test_run_signals.py` | `reporting/run_signals` (incr/pop consume + per-video isolation;
emit consumes signals even with reporting OFF). ? deliberately NO obsreport dep ? this is the
half of the oversize channel CI can verify (obsreport is absent there) |
331:| `test_reporting_spec_faq.py` | `reporting/spec.yaml` <-> `README.md` cross-ref integrity
(every `faq_ref` resolves; actionable rules cite a `faq_ref`); ? `importorskip("obsreport")` |

```

14. user (2026-06-12T14:57:46.498Z)

```

docs\FIELD_VISIT_ANSWER_SHEET.md:4:> is a *good* sheet. Companion script:
`FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
docs\FIELD_VISIT_ANSWER_SHEET.md:5:> Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` §2.
docs\FIELD_VISIT_ANSWER_SHEET.md:78:## Scores ? OFFICE USE, leave blank (scored by Avi at
transcription per PMF_FIELD_SIGNALS.md §2; when torn, score lower)

```

docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md:3:> ****Recruiting?**** Hand candidates
 [`PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md`](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
 docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md:66:****After each visit, fill ONE answer sheet ? the
 template is [`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md)**
 docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md:70:[`PMF\_FIELD\_SIGNALS.md`](PMF_FIELD_SIGNALS.md) ? that
 part is ours, not yours; just capture honestly.)*
 docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md:114:([`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) ? one per academy).
 docs\NEXT_STEPS.md:61: `docs/PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md` (the recruiting flyer ? owner
 fills ?/contact placeholders and recruits) +
 docs\NEXT_STEPS.md:62: `docs/FIELD\_RESEARCH\_ASSOCIATE\_FLYER.md` (the hired associate's visit
 playbook: coach + student script, demo-after-Q6
 docs\NEXT_STEPS.md:63: rule, ground rules, Phase-2 record-and-show-back stretch goal) +
 `docs/FIELD\_VISIT\_ANSWER\_SHEET.md` (per-visit
 docs\NEXT_STEPS.md:64: capture form incl. demo-reaction + student blocks) +
 `docs/PMF\_FIELD\_SIGNALS.md` (G/A/R anchors + ****pre-registered**
 docs\PMF_FIELD_SIGNALS.md:6:****Companions:****
 [`PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md`](PRODUCT_FIELD_ASSOCIATE_FLYER.md)
 docs\PMF_FIELD_SIGNALS.md:8:[`FIELD\_RESEARCH\_ASSOCIATE\_FLYER.md`](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
 docs\PMF_FIELD_SIGNALS.md:10:[`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) (the
 per-visit capture form) .
 docs\PMF_FIELD_SIGNALS.md:122:- One [`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md)
 per academy,

15. user (2026-06-12T14:57:56.504Z)

2-
 3:****Status:**** PROPOSAL ? awaiting owner ratification (no implementation starts without explicit
 approval, per CLAUDE.md).
 4:****Date:**** 2026-06-10 . ****Inputs:**** PMF_MARKET_RESEARCH, COMMERCIALIZATION,
 PLATFORM_ARCHITECTURE, DEFERRED_HARDENING_PLAN (#16), ROADMAP, VIDEO_RECORDING_GUIDELINES,
 reporting/spec.yaml, current API + static UI.
 5:****Companion:**** `docs/HOSTING\_PLAN.md` (domain, topology, costs).
 6-
 --
 43-| A1 | ****Matches**** (home) | User's videos, status chip per video (verdict ?/??/?), upload CTA
 | `videos` + owner scoping (B4) |
 44-| A2 | ****Upload & Process**** | Drag-drop; inline recording-guidelines checklist (pre-upload
 education from VIDEO_RECORDING_GUIDELINES); chunked upload; then auto-starts processing | B2
 upload endpoint ? B1 job |
 45-| A3 | ****Job progress**** | Stage list (validation ? segmentation ? AI confirm ? report) with
 live status; verdict banner on completion | B1 `GET /api/jobs/{id}` poll |
 46-| A4 | ****Rally review**** ? | Video player + rally list; click rally ? seek & play that span;
 per-rally ****Accept \ Reject \ nudge start?**end ± 0.5 s******; bulk "all good"; chips: duration, shot
 count (+confidence) | B6 correction endpoints ? `candidate\_segments.human\_label/notes` ? ****the
 corpus engine**** |
 47-| A5 | ****Highlights**** | Select accepted rallies ? name reel ? compile ? download; list of
 past reels | `/api/compile` + B3 download endpoint |
 48-| A6 | ****Report**** | Renders the customer-audience report (verdict, notes, FAQ refs); "nerd
 view" toggle shows technical.md | existing reporting artifacts |
 49-
 50-### Backend work the alpha actually requires (honest dependency list)

16. user (2026-06-12T14:57:57.397Z)

40 Priority order for new videos: ******(a) multi-court badminton****** (the target distribution
 AND where the ship target
 41 lives), ******(b) ?3 matches per new sport ? table tennis first****** (WASB transfers at F1
 0.909; pickleball labels are
 42 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
 capture variety per
 43 `VIDEO\_RECORDING\_GUIDELINES.md`; ****adult sessions only for the training pool****
 (consent guardrail). Per video:
 44 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?

re-run the Gen-0 harness.

45 The paired sign test needs n: at n=10 with 9 wins, $p \approx 0.011$? significance is reachable this month.

46 2. ****[owner decision] Set the ship-gate target**** ? now informed: floor = heuristic 0.480 (necessary), ****multi-court**

47 reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93

48 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with

49 paired-sign $p < 0.05$ vs heuristic" ? owner to ratify/adjust.

50 3. ****M0.4 next increment (\$0/local):**** swap the Gen-0 LogReg for ****ActionFormer/ASFormer**** (MIT, minutes on the

51 2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold

52 calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).

53 4. ****Multisport thread:**** ingest Roora pickleball/TT CSVs (``import-local --normalize``); merged-prototype LOVO across

54 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated

55 by C3 (provenance multiplies per sport).

56 5. ****PMF validation (cheaper than any engineering month):**** C13 academy interviews (+ the two added questions:

57 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies

58 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:

59 `output/MahadevpuraSingles\_highlights.mp4`). ****Field kit COMPLETE (2026-06-12, owner ratified the**

60 physical-visit approach; reframed product-feedback-first per owner spec):**

61 `docs/PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +

62 `docs/FIELD\_RESEARCH\_ASSOCIATE\_FLYER.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6

63 rule, ground rules, Phase-2 record-and-show-back stretch goal) +

64 `docs/FIELD\_VISIT\_ANSWER\_SHEET.md` (per-visit capture form incl. demo-reaction + student blocks) + `docs/PMF\_FIELD\_SIGNALS.md` (G/A/R anchors + ****pre-registered**

65 decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).

66 6. ****Housekeeping:**** ~~rotate the Gemini API key~~ ****DONE 2026-06-10**** (owner-verified working; confirm old-key

67 deletion in AI Studio); ****rename repo off "badminton"**** (owner action,

68 in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery completed.

69

70 ## Pointers / artifacts

71 - Roadmap: `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` · Topology: ADR-008 (``docs/archives/decisions/DECISIONS.md``) ·

72 Study: `docs/archives/research/OWNED\_MODEL\_TRAINING\_STUDY.md` · Quality history: ``docs/archives/quality-iterations/2026-06-multivideo-lovo/``.

73 - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle; weights gitignored,

74 manifest reviewable). Featurizer: `backend/features/rally\_seq.py` (FEATURE_SCHEMA v1).

75 - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest ``output/human_lovo_manifest.json``

76 (gitignored) · durable trajectories + Roora files: ``C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\`` ·

77 labels: ``...\Collect\Golden Labelled\``.

78 - Gemini ops learnings: serial uploads (``indexing.ai_max_workers=1``) + chunk re-encode

79 (``indexing.ai_chunk_scale_width=1280``) + 6-attempt exponential generate retry ? a cold/prepaid project burst-throttles

80 with a MISLEADING "credits depleted" 429; ramp gradually.

81 - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next milestone. Committed next

82 PLATFORM slice = async job model (single-tenant).

83

17. user (2026-06-12T14:58:53.672Z)

```

1      # Badminton Highlight Indexer ? Video Report (summary)
2
3      ## Run
4
5      | Field | Value |
6      | --- | --- |
7      | Video | kushagra_1280.mp4 |
8      | Report generated | 2026-06-10T11:31:17Z |
9
10     ## Video
11
12     | Field | Value |
13     | --- | --- |
14     | Duration | 8m58s |
15     | Resolution | 1280x720 |
16     | Frame rate | 59.94 fps |
17
18     ## Highlights
19
20     - **13** highlights
21     - **57.0s** of play
22     - **10.6%** of the video is highlights
23
24     ## Things to know
25
26     _Nothing needs your attention._
27
28     ## Verdict
29
30     ? Your highlights are ready.
31

```

18. user (2026-06-12T15:00:58.958Z)

Structured output provided successfully

19. assistant (2026-06-12T15:01:08.005Z)

Findings delivered via StructuredOutput ? 10 findings total: 1 high (the "private web app is opening" present-tense claim, which both overstates the unratified hosting/unbuilt SPA state and pre-asserts the PROCEED verdict the campaign exists to decide), 3 medium (the nonexistent "match report" baked into the H5 pricing question and Phase-2 show-back; the "senior or adult" minors loophole in the student-interview flow; the associate-facing links into the scoring/economics layer that undermine the pre-registration anchoring design), and 6 low (15-vs-30-minute opening promise, flyer/playbook pay and paired-visit mismatches, undefined "completed visit", "finds every rally" absolutism, bundled Phase-2 consent vs the separate training opt-in guardrail, and stale "flyer rule N" references in PMF_FIELD_SIGNALS after the reframe). Verified consistent: the demo-after-Q6 rule across all four docs, the associate-carries-the-demo framing, and the minors recording exclusion.